PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 4

Quadratic Equations

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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❓ In-text questions (Why?)

Example 1 (Why?) **Let John have x marbles. Then why does Jivanti have 45 − x marbles?**

✔ ANSWER

Together they have 45 marbles. If John has x, the rest belong to Jivanti.

So Jivanti has **45 − x** marbles.

Example 8 (Why?) **AB is a diameter of the circular park and P is a point on the boundary. Why is ∠APB = 90°?**

✔ ANSWER

The angle in a semicircle is a right angle (an angle subtended by a diameter at any point on the circle is 90°).

So **∠APB = 90°**, and Pythagoras theorem can be used in ΔAPB.

📝 Exercise 4.1

1 (i) **Check whether $(x + 1)^2 = 2(x - 3)$ is a quadratic equation.**

✔ ANSWER

LHS = $x^2 + 2x + 1$, RHS = $2x - 6$

$x^2 + 2x + 1 = 2x - 6$

$x^2 + 7 = 0$ → of the form $ax^2 + bx + c = 0$ ($a = 1$, $b = 0$, $c = 7$)

Yes, it is a quadratic equation.

1 (ii) **Check whether $x^2 - 2x = (-2)(3 - x)$ is a quadratic equation.**

✔ ANSWER

RHS = $-6 + 2x$

$x^2 - 2x = 2x - 6$

$x^2 - 4x + 6 = 0$ → of the form $ax^2 + bx + c = 0$

Yes, it is a quadratic equation.

1 (iii) **Check whether $(x - 2)(x + 1) = (x - 1)(x + 3)$ is a quadratic equation.**

✔ ANSWER

LHS = $x^2 + x - 2x - 2 = x^2 - x - 2$

RHS = $x^2 + 3x - x - 3 = x^2 + 2x - 3$

$x^2 - x - 2 = x^2 + 2x - 3$

$-3x + 1 = 0$ → the x^2 terms cancel; degree is 1

No, it is not a quadratic equation.

1 (iv) **Check whether $(x - 3)(2x + 1) = x(x + 5)$ is a quadratic equation.**

✔ ANSWER

LHS = $2x^2 + x - 6x - 3 = 2x^2 - 5x - 3$

RHS = $x^2 + 5x$

$2x^2 - 5x - 3 = x^2 + 5x$

$x^2 - 10x - 3 = 0$ → of the form $ax^2 + bx + c = 0$

Yes, it is a quadratic equation.

1 (v) **Check whether $(2x - 1)(x - 3) = (x + 5)(x - 1)$ is a quadratic equation.**

✔ ANSWER

LHS = $2x^2 - 6x - x + 3 = 2x^2 - 7x + 3$

RHS = $x^2 - x + 5x - 5 = x^2 + 4x - 5$

$2x^2 - 7x + 3 = x^2 + 4x - 5$

$x^2 - 11x + 8 = 0$ → of the form $ax^2 + bx + c = 0$

Yes, it is a quadratic equation.

1 (vi) **Check whether $x^2 + 3x + 1 = (x - 2)^2$ is a quadratic equation.**

✔ ANSWER

RHS = $x^2 - 4x + 4$

$x^2 + 3x + 1 = x^2 - 4x + 4$

$7x - 3 = 0$ → the x^2 terms cancel; degree is 1

No, it is not a quadratic equation.

1 (vii) **Check whether $(x + 2)^3 = 2x(x^2 - 1)$ is a quadratic equation.**

✔ ANSWER

LHS = $x^3 + 6x^2 + 12x + 8$ [$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$]

RHS = $2x^3 - 2x$

$x^3 + 6x^2 + 12x + 8 = 2x^3 - 2x$

$-x^3 + 6x^2 + 14x + 8 = 0$, i.e. $x^3 - 6x^2 - 14x - 8 = 0$ → degree 3 (cubic)

No, it is not a quadratic equation.

1 (viii) **Check whether $x^3 - 4x^2 - x + 1 = (x - 2)^3$ is a quadratic equation.**

✔ ANSWER

RHS = $x^3 - 6x^2 + 12x - 8$

$x^3 - 4x^2 - x + 1 = x^3 - 6x^2 + 12x - 8$

$2x^2 - 13x + 9 = 0$ → the x^3 terms cancel; of the form $ax^2 + bx + c = 0$

Yes, it is a quadratic equation.

2 (i) **The area of a rectangular plot is 528 m^2. The length (in metres) is one more than twice its breadth. Represent this as a quadratic equation (to find the length and breadth).**

✔ ANSWER

Let the breadth be x m. Then length = (2x + 1) m.

Area = length × breadth = $(2x + 1)x = 528$

$2x^2 + x = 528$

$2x^2 + x - 528 = 0$, where x is the breadth of the plot in metres.

2 (ii) **The product of two consecutive positive integers is 306. Represent this as a quadratic equation (to find the integers).**

✔ ANSWER

Let the smaller integer be x. The next integer = x + 1.

$x(x + 1) = 306$

$x^2 + x = 306$

$x^2 + x - 306 = 0$, where x is the smaller integer.

2 (iii) **Rohan's mother is 26 years older than him. The product of their ages (in years) 3 years from now will be 360. Represent this as a quadratic equation (to find Rohan's present age).**

✔ ANSWER

Let Rohan's present age be x years. Mother's present age = (x + 26) years.

After 3 years: Rohan = x + 3, mother = x + 29

$(x + 3)(x + 29) = 360$

$x^2 + 29x + 3x + 87 = 360$

$x^2 + 32x + 87 - 360 = 0$

$x^2 + 32x - 273 = 0$, where x is Rohan's present age in years.

2 (iv) **A train travels 480 km at a uniform speed. If the speed had been 8 km/h less, it would have taken 3 hours more to cover the same**

distance. Represent this as a quadratic equation (to find the speed of the train).

✔ ANSWER

Let the speed be u km/h. Time taken = 480/u hours.

With speed (u − 8) km/h, time = 480/(u − 8) hours, which is 3 hours more:

$480/(u - 8) - 480/u = 3$

$480[u - (u - 8)] = 3u(u - 8)$

$480 \times 8 = 3u^2 - 24u$

$3840 = 3u^2 - 24u \rightarrow$ divide by 3: $1280 = u^2 - 8u$

$u^2 - 8u - 1280 = 0$, where u is the speed of the train in km/h.

📝 Exercise 4.2

1 (i) **Find the roots of $x^2 - 3x - 10 = 0$ by factorisation.**

✔ ANSWER

Split the middle term: $-3x = -5x + 2x$ [$(-5) \times 2 = -10$, $-5 + 2 = -3$]

$x^2 - 5x + 2x - 10 = 0$

$x(x - 5) + 2(x - 5) = 0$

$(x - 5)(x + 2) = 0$

$x - 5 = 0$ or $x + 2 = 0$

Roots: $x = 5, x = -2$

1 (ii) **Find the roots of $2x^2 + x - 6 = 0$ by factorisation.**

✔ ANSWER

Split the middle term: $x = 4x - 3x$ [$4 \times (-3) = -12 = 2 \times (-6)$]

$2x^2 + 4x - 3x - 6 = 0$

$2x(x + 2) - 3(x + 2) = 0$

$(x + 2)(2x - 3) = 0$

$x = -2$ or $x = 3/2$

Roots: $x = -2, x = 3/2$

1 (iii) **Find the roots of $\sqrt{2}\,x^2 + 7x + 5\sqrt{2} = 0$ by factorisation.**

✔ ANSWER

Product of coefficients: $\sqrt{2} \times 5\sqrt{2} = 10$. Split $7x = 5x + 2x$ ($5 \times 2 = 10$).

$\sqrt{2}\,x^2 + 5x + 2x + 5\sqrt{2} = 0$

$x(\sqrt{2}\,x + 5) + \sqrt{2}(\sqrt{2}\,x + 5) = 0$ [since $\sqrt{2} \times \sqrt{2}\,x = 2x$ and $\sqrt{2} \times 5 = 5\sqrt{2}$]

$(\sqrt{2}\,x + 5)(x + \sqrt{2}) = 0$

$\sqrt{2}\,x + 5 = 0 \rightarrow x = -5/\sqrt{2}$; $x + \sqrt{2} = 0 \rightarrow x = -\sqrt{2}$

Roots: $x = -5/\sqrt{2}, x = -\sqrt{2}$

1 (iv) **Find the roots of $2x^2 - x + 1/8 = 0$ by factorisation.**

✔ ANSWER

Multiply by 8: $16x^2 - 8x + 1 = 0$

Split $-8x = -4x - 4x$ [$(-4) \times (-4) = 16 = 16 \times 1$]

$16x^2 - 4x - 4x + 1 = 0$

$4x(4x - 1) - 1(4x - 1) = 0$

$(4x - 1)(4x - 1) = 0$

$4x - 1 = 0 \rightarrow x = 1/4$ (repeated)

Roots: x = 1/4, 1/4 (two equal roots)

1 (v) **Find the roots of $100x^2 - 20x + 1 = 0$ by factorisation.**

✔ ANSWER

Split $-20x = -10x - 10x$ [$(-10) \times (-10) = 100 = 100 \times 1$]

$100x^2 - 10x - 10x + 1 = 0$

$10x(10x - 1) - 1(10x - 1) = 0$

$(10x - 1)(10x - 1) = 0$

$10x - 1 = 0 \rightarrow x = 1/10$ (repeated)

Roots: x = 1/10, 1/10 (two equal roots)

2 (i) **Solve the problem of Example 1 (i): John and Jivanti together have 45 marbles. Both lost 5 marbles each, and the product of the marbles they now have is 124. How many marbles did each have to start with?**

✔ ANSWER

From Example 1, John's marbles x satisfy $x^2 - 45x + 324 = 0$

Split $-45x = -36x - 9x$ [$(-36) \times (-9) = 324$, $-36 - 9 = -45$]

$x^2 - 36x - 9x + 324 = 0$

$x(x - 36) - 9(x - 36) = 0$

$(x - 36)(x - 9) = 0 \rightarrow x = 36$ or $x = 9$

If John had 36, Jivanti had $45 - 36 = 9$; if John had 9, Jivanti had 36.

Check: $(36 - 5)(9 - 5) = 31 \times 4 = 124$ ✓

They had 36 and 9 marbles (John 36 and Jivanti 9, or John 9 and Jivanti 36).

2 (ii) **Solve the problem of Example 1 (ii): the cost of each toy (in ₹) was 55 minus the number of toys produced in a day. The total cost of production that day was ₹750. Find the number of toys produced.**

✔ ANSWER

From Example 1, the number of toys x satisfies $x^2 - 55x + 750 = 0$

Split $-55x = -25x - 30x$ [$(-25) \times (-30) = 750$, $-25 - 30 = -55$]

$x^2 - 25x - 30x + 750 = 0$

$x(x - 25) - 30(x - 25) = 0$

$(x - 25)(x - 30) = 0 \rightarrow x = 25$ or $x = 30$

Check: 25 toys at ₹$(55 - 25)$ = ₹30 each → ₹750 ✓; 30 toys at ₹25 each → ₹750 ✓

Number of toys produced = 25 or 30

3 **Find two numbers whose sum is 27 and product is 182.**

✔ ANSWER

Let one number be x. The other = 27 − x.

$x(27 - x) = 182$

$27x - x^2 = 182$

$x^2 - 27x + 182 = 0$

Split $-27x = -13x - 14x$ [$(-13) \times (-14) = 182$]

$x(x - 13) - 14(x - 13) = 0$

$(x - 13)(x - 14) = 0 \rightarrow x = 13$ or $x = 14$

If x = 13, other = 14; if x = 14, other = 13.

The numbers are 13 and 14.

4 **Find two consecutive positive integers, sum of whose squares is 365.**

✔ ANSWER

Let the integers be x and x + 1.

$x^2 + (x + 1)^2 = 365$

$x^2 + x^2 + 2x + 1 = 365$

$2x^2 + 2x - 364 = 0 \rightarrow$ divide by 2: $x^2 + x - 182 = 0$

Split $x = 14x - 13x$ [$14 \times (-13) = -182$]

$x(x + 14) - 13(x + 14) = 0$

$(x + 14)(x - 13) = 0 \rightarrow x = -14$ or $x = 13$

x is positive, so x = 13 and x + 1 = 14.

Check: $13^2 + 14^2 = 169 + 196 = 365$ ✓

The integers are 13 and 14.

5 **The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.**

✔ ANSWER

Let the base be x cm. Then altitude = (x − 7) cm.

By Pythagoras theorem: $x^2 + (x - 7)^2 = 13^2$

$x^2 + x^2 - 14x + 49 = 169$

$2x^2 - 14x - 120 = 0$ → divide by 2: $x^2 - 7x - 60 = 0$

Split $-7x = -12x + 5x$ $[(-12) \times 5 = -60]$

$x(x - 12) + 5(x - 12) = 0$

$(x - 12)(x + 5) = 0$ → $x = 12$ or $x = -5$

A side cannot be negative, so x = 12.

Base = 12 cm, altitude = 12 − 7 = 5 cm. Check: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ ✓

The other two sides are 12 cm and 5 cm.

6 **A cottage industry produces some pottery articles in a day. The cost of production of each article (in ₹) was 3 more than twice the number of articles produced that day. If the total cost of production was ₹90, find the number of articles produced and the cost of each article.**

✔ ANSWER

Let the number of articles be x. Cost of each article = ₹(2x + 3).

Total cost = $x(2x + 3) = 90$

$2x^2 + 3x - 90 = 0$

Split $3x = 15x - 12x$ [$15 \times (-12) = -180 = 2 \times (-90)$]

$2x^2 + 15x - 12x - 90 = 0$

$x(2x + 15) - 6(2x + 15) = 0$

$(2x + 15)(x - 6) = 0 \rightarrow x = -15/2$ or $x = 6$

Number of articles cannot be negative, so x = 6.

Cost of each article = 2 × 6 + 3 = ₹15. Check: 6 × 15 = 90 ✓

Number of articles = 6, cost of each article = ₹15

Exercise 4.3

1 (i) **Find the nature of the roots of $2x^2 - 3x + 5 = 0$. If real roots exist, find them.**

✔ ANSWER

a = 2, b = −3, c = 5

Discriminant $D = b^2 - 4ac = (-3)^2 - 4 \times 2 \times 5 = 9 - 40 = -31$

$D < 0$

No real roots exist.

1 (ii) **Find the nature of the roots of $3x^2 - 4\sqrt{3}x + 4 = 0$. If real roots exist, find them.**

✔ ANSWER

$a = 3, b = -4\sqrt{3}, c = 4$

$D = b^2 - 4ac = (-4\sqrt{3})^2 - 4 \times 3 \times 4 = 48 - 48 = 0$

$D = 0$, so the roots are real and equal.

Each root $= -b/2a = 4\sqrt{3}/(2 \times 3) = 4\sqrt{3}/6 = 2\sqrt{3}/3 = 2/\sqrt{3}$

Two equal real roots: $2/\sqrt{3}, 2/\sqrt{3}$

1 (iii) **Find the nature of the roots of $2x^2 - 6x + 3 = 0$. If real roots exist, find them.**

✔ ANSWER

$a = 2, b = -6, c = 3$

$D = b^2 - 4ac = (-6)^2 - 4 \times 2 \times 3 = 36 - 24 = 12$

$D > 0$, so there are two distinct real roots.

$x = [-b \pm \sqrt{D}]/2a = [6 \pm \sqrt{12}]/4 = [6 \pm 2\sqrt{3}]/4 = (3 \pm \sqrt{3})/2$

Roots: $(3 + \sqrt{3})/2$ and $(3 - \sqrt{3})/2$

2 (i) **Find the value of k so that $2x^2 + kx + 3 = 0$ has two equal roots.**

✔ ANSWER

$a = 2, b = k, c = 3$

For equal roots, $D = 0$: $k^2 - 4 \times 2 \times 3 = 0$

$k^2 - 24 = 0 \rightarrow k^2 = 24$

$k = \pm\sqrt{24} = \pm 2\sqrt{6}$

$k = 2\sqrt{6}$ or $k = -2\sqrt{6}$

2 (ii) **Find the value of k so that $kx(x - 2) + 6 = 0$ has two equal roots.**

✔ ANSWER

$kx^2 - 2kx + 6 = 0 \rightarrow a = k, b = -2k, c = 6$ ($k \neq 0$, otherwise it is not quadratic)

For equal roots, $D = 0$: $(-2k)^2 - 4 \times k \times 6 = 0$

$4k^2 - 24k = 0$

$4k(k - 6) = 0 \rightarrow k = 0$ or $k = 6$

$k = 0$ is not allowed (the equation would not be quadratic).

$k = 6$

3 **Is it possible to design a rectangular mango grove whose length is twice its breadth, and the area is 800 m²? If so, find its length and breadth.**

✔ ANSWER

Let the breadth be x m. Then length = 2x m.

Area: $2x \times x = 800 \rightarrow 2x^2 = 800$

$x^2 = 400 \rightarrow x^2 - 400 = 0$ ($a = 1, b = 0, c = -400$)

$D = 0^2 - 4 \times 1 \times (-400) = 1600 > 0$, so real roots exist; it is possible.

$x = \pm 20$. Breadth cannot be negative, so $x = 20$.

Length = 2 × 20 = 40 m. Check: 40 × 20 = 800 ✓

Yes. Length = 40 m, breadth = 20 m

4 **Is the following situation possible? If so, find their present ages. The sum of the ages of two friends is 20 years. Four years ago, the product of their ages in years was 48.**

✔ ANSWER

Let one friend's age be x years. The other's age = (20 − x) years.

Four years ago: (x − 4) and (20 − x − 4) = (16 − x)

$(x - 4)(16 - x) = 48$

$16x - x^2 - 64 + 4x = 48$

$-x^2 + 20x - 112 = 0 \rightarrow x^2 - 20x + 112 = 0$

$D = (-20)^2 - 4 \times 1 \times 112 = 400 - 448 = -48$

$D < 0$, so there are no real roots.

No, the situation is not possible.

5 **Is it possible to design a rectangular park of perimeter 80 m and area 400 m²? If so, find its length and breadth.**

✔ ANSWER

Let the length be x m. Perimeter $2(l + b) = 80$ → $l + b = 40$, so breadth = (40 − x) m.

Area: $x(40 - x) = 400$

$40x - x^2 = 400 \rightarrow x^2 - 40x + 400 = 0$

$D = (-40)^2 - 4 \times 1 \times 400 = 1600 - 1600 = 0$

$D = 0$, so real (equal) roots exist; it is possible.

$x = -b/2a = 40/2 = 20$. Breadth $= 40 - 20 = 20$ m.

Check: perimeter $2(20 + 20) = 80$ ✓, area $20 \times 20 = 400$ ✓

Yes. Length = 20 m, breadth = 20 m (the park is a square)

🎉 **Chapter 4 done! Revise the answers once more tomorrow.**